

Perplexity

- * evaluation metric

=)

e raised to cross entropy loss

- * perplexity of 10 means

model on average as uncertain as if it was choosing uniformly between 10 likely (equally) tokens.

- * Why perplexity for compression.

Run model on a long document with full KV cache

↳ measure perplexity $\rightarrow P_a$

then apply compression techniques (eviction, quantisation)

↳ measure perplexity $\rightarrow P_b$

$$P_b - P_a$$

if small gap $\rightarrow$ compression is working

Needle in Haystack

perplexity is an average metric

but completely fail on one critical schri

Needle in haystack catches this.

Long document $\rightarrow$ specific fact @ some random pt. (the haystack)

@ the end ask it to retrieve the fact

we sweep

- a) depth $\rightarrow$ where it is placed
- b) document length.

produce a 2D heatmap: depth $\times$ length

it connects to dormant tokens

(exactly a dormant token) $\rightarrow$ near 0 attn.

Quality vs Memory Curves

Sweep cache budget: keep 100%, 80%, 60%, of full cache

& @ each budget level run

perplexity & needle in a haystack

& plot two curves

perplexity vs cache

retrieval accuracy vs cache

Multi Latent Attention.

Compresses K & V into a smaller vector by
caching & decompress.

$$C_t = h_t \cdot W_{\text{down}}$$

has dim $d_c \ll d_k$ (512 vs 16384)

only C_t gets cached

& during inference & attention input-

$$K = C_t \cdot W_{\text{up}K}$$

$$V = C_t \cdot W_{\text{up}V}$$

Diff from GQA $\rightarrow$ shares full size
 K/V vectors across

heads,

BASICALLY COMPRESSED ZIP FILE

